

LEAVING NO ONE BEHIND: PRIORITISING DIGITAL HEALTH EQUITY ACROSS THE WHO EUROPEAN REGION

By: Keyrellous Adib, Natasha Azzopardi-Muscat and David Novillo-Ortiz

Summary: The 2023 World Health Organization (WHO) report on digital health for the WHO European Region demonstrates the accelerated progress in the adoption of digital health technologies (DHT) in recent years. However, there are growing concerns about the digital divide resulting from inequitable access and utilisation of DHT, particularly among older people and marginalised communities. This article sheds light on the advancements made by WHO Europe Member States and underscores the necessity to ensure access to devices and stable internet connection, as well as promoting digital literacy and user engagement, to overcome this digital divide.

Keywords: Digital Health Equity, Digital Divide, Digital Health Inclusion, Digital Health Literacy

> #EPH2023 – PLENARY 4:

On the path to an equitable and sustainable digital future for European health systems

Keyrellous Adib is Technical Officer for Data Science and Digital Health, Data and Digital Health team, Division of Country Health Policies and Systems, **Natasha Azzopardi-Muscat** is Director, Division of Country Health Policies and Systems, **David Novillo-Ortiz** is Regional Advisor for Data and Digital Health, Division of Country Health Policies and Systems, World Health Organization, Regional Office for Europe, Copenhagen, Denmark. Email: dnovillo@who.int

Introduction

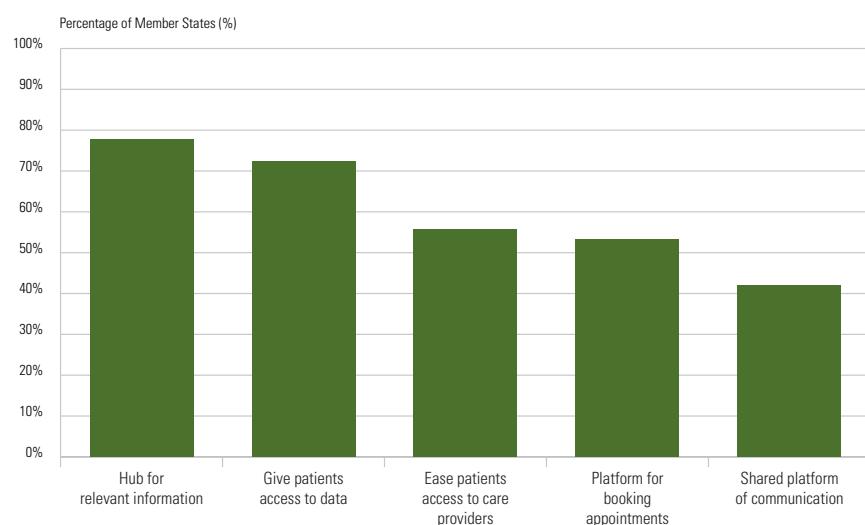
As the integration of digital health technologies (DHT) within the healthcare system continues to expand, it's crucial to prioritise equitable access and utilisation. This goes beyond just ensuring the presence of suitable devices and stable internet connection; it also necessitates improving user engagement and promoting digital health literacy, enabling individuals to effectively use technology to access and understand health information.¹

Transforming patient-provider engagement in healthcare

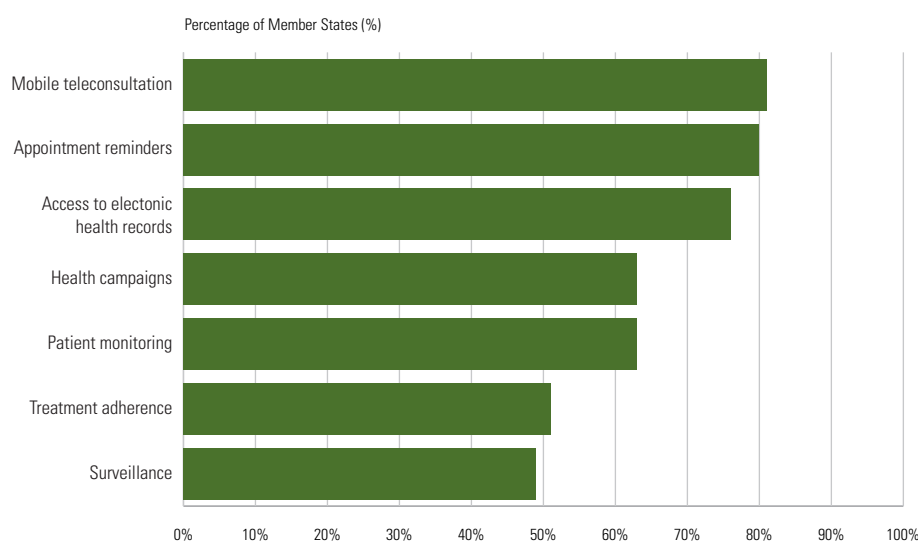
The rise in adoption of DHT is bringing numerous advantages and revolutionising the way patients and healthcare providers

engage with the healthcare system. For instance, patient portals, which are online platforms, grant patients secure and real-time access to their personal health records from anywhere with an internet connection. Within the WHO European Region*, 71% of Member States have established a national digital health portal.² Among these Member States (see Figure 1), 78% utilise the patient portal primarily as an information centre, 72% enable patients to retrieve their health records through the portal, 53% have integrated functionalities for appointment booking, 56% have developed these

* All 53 countries of the WHO European region took part in the survey on which the report on digital health for the WHO European Region is based. However, responses to questions in the survey were not mandatory and so response rates were not uniform across all questions.

Figure 1: Uses of National Digital Health Portals in the WHO European Region

Source: [1]

Figure 2: Types of mHealth programmes and services for healthcare in the WHO European Region

Source: [1]

portals to ensure that patient information is readily available to healthcare providers, and 42% have embedded communication features into their portals.^[1]

Telehealth has the potential to bridge geographical and socio-economic barriers, ensuring that individuals in remote or underserved areas have access to quality healthcare services. In the WHO European Region, 84% of Member States reported using teleradiology.^[1] Additionally, 77% utilised telemedicine or remote patient monitoring. More than half of the Member

States reported using teleradiology and telepsychiatry.^[1] However, telehealth services' distribution varies across the WHO European Region, with northern and western Europe showcasing a broad range of telehealth services, whereas western Asia and central Asia have a more limited range of services.^[1] Teleradiology is notably prevalent, with over 80% of Member States in all subregions, excluding central Asia, adopting this service.^[1] Telemedicine or remote patient monitoring is highly used in southern Europe,

Europe, and western Europe by 90%, 91% and 100% of Member States, respectively.^[1] In stark contrast, only 56% of the Member States in eastern Europe, and 25% of Member States in central Asia reported having telemedicine services.^[1]

Mobile health (mHealth) services and applications leverage the capabilities of smartphones, transforming the healthcare landscape by facilitating immediate access, enhancing personal engagement, and streamlining remote healthcare services. Between 2015 and 2022, there was a noticeable increase in the number of Member States that reported having at least one government-backed mHealth service.^[1] The percentage of Member States with such services surged from 49% in 2015 to 91% by 2022.^[1] As shown in **Figure 2**, teleconsultations is the most prevalent mHealth service available in 81% of the Member States, followed closely by appointment reminders in 80% of the Member States, and access to electronic patient information from electronic health records in 76% of the Member States. Treatment adherence services, which can help increase patient compliance with medical regimens, were reported by over half of the Member States.^[1] Additionally, 63% of Member States have reported having mHealth services for health promotion, community mobilisation, and risk communication. The same percentage also reported having mHealth programmes for patient monitoring, facilitating the capture and transmission of data for diverse medical conditions. Surveillance, an essential component for data collection, management, and reporting in healthcare, is facilitated through mobile devices in 49% of Member States.^[1]

Demographics of the digital divide

Despite advances in digital transformation by the Member States of the WHO European Region, concerns arise regarding equitable access to DHT. WHO's recent scoping review provided consistent evidence indicating that there is a higher utilisation of DHT in urban areas compared to rural ones.^[2] Moreover, individuals from ethnic minorities and those facing language barriers have lower DHT usage.^[2] The review also found higher DHT use by individuals

Table 1: Most cited measures to ensure equity in accessing digital health services

MEASURE	NUMBER OF MEMBER STATES
Improvements in connectivity (broadband) and ICT infrastructure	22
Digital literacy and digital health promotion	21
Expanding geographical coverage and the range of care services using telemedicine solutions	19
Improving patient access to digital health data and health information	14
Assistance to vulnerable populations (aid to the elderly/physically impaired etc.)	5
Development of guidelines	1

Source: ¹

with higher educational levels, those of a higher economic status, and younger individuals. Furthermore, better access to DHT was prevalent among individuals without disabilities or complex health needs.² Additionally, the review pointed out that there is limited evidence on how DHT helps improve the health and well-being of groups that are often left out, like the homeless or those with substance use disorders.³

Eurostat data further highlight this digital divide. In the European Union (EU), 81% of urban residents access the internet daily, compared to 70% in rural areas.⁴ Those employed exhibit superior digital skills than the unemployed.⁵ Students stand out with 68.2% having more than basic digital skills.⁶ Moreover, there is a salient disparity in internet connectivity among households based on income. While 99% of households in the highest income quartile have internet access, only 73% in the poorest quartile do.⁷ Additionally, a striking difference in digital literacy exists between age groups: only 33% of those aged 55-74 years possess basic digital skills, compared to 80% of individuals aged 16-24 years.⁸

Digital health strategies and measures of equity

Digital health governance plays a vital role in promoting universal health coverage (UHC), emphasising inclusive healthcare for all. In the WHO European Region, 83% of Member States reported having a digital health strategy or similar policy.⁹ Notably, 71% of Member States

have explicitly listed UHC as one of the strategic priorities of their digital health policies and strategies.¹⁰ It is also noteworthy that almost all the responding Member States, except one, highlighted enhancing accessibility, quality, safety, and efficiency as their strategic priorities. These priorities align directly with the essential elements of UHC.¹¹

Recognising the adverse consequences resulting from the digital divide, 75% of Member States with digital health strategies or similar policies, reported that their policies or strategies included measures to ensure equity in accessing digital healthcare services.¹² The majority of these Member States identified two measures: improving digital literacy among general public and health professionals and ensuring reliable internet connectivity. As shown in **Table 1**, 21 Member States prioritised enhancing digital literacy, while 22 Member States emphasised the importance of improving internet connectivity and ICT infrastructure.¹³ These priorities underscore the critical challenges faced by Member States in bridging the digital divide in healthcare: ensuring not only the availability of technology but also equipping their populations with the skills to utilise it effectively.¹⁴

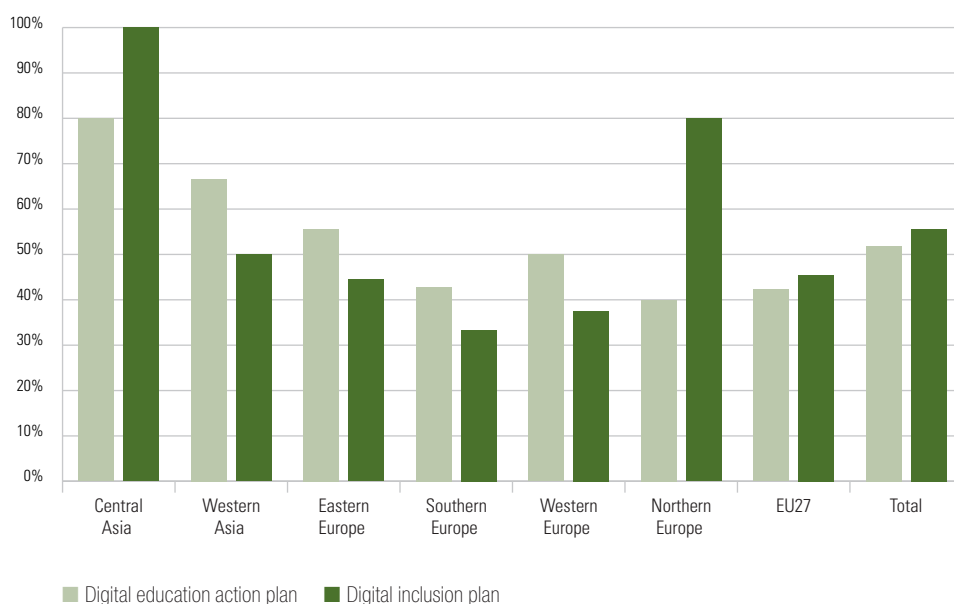
Digital health literacy and inclusion plans

Digital health literacy refers to the capacity to seek, locate, comprehend, and evaluate electronic health information and apply the knowledge to address health concerns.¹⁵

WHO/Europe Resolution (EUR/RC69/R9) on health literacy that was adopted at the WHO Regional Committee for Europe in 2019, put emphasis on DHT and digital health literacy, and improving digital health literacy is one of the strategic areas of the Draft WHO European roadmap for implementation of health literacy initiatives through the life course.¹⁶

Digital health literacy encompasses a variety of individual and societal factors, as well as technological constraints, that influence people's ability to effectively use digital technologies for searching, acquiring, understanding, appraising, and communicating health-related information to maintain or improve their health and quality of life.¹⁷ Therefore, promoting digital health literacy in the general population is critical to ensuring health equity and access to high-quality healthcare services and information for all. Low levels of digital health literacy can lead to misinformation and confusion, resulting in poor health management, delayed treatments, and higher healthcare costs.

Plans and strategies for digital health education and digital inclusion are essential to ensure that everyone, regardless of background, has access to and can effectively use digital technologies. Digital inclusion plans, policies and strategies aim to bridge the digital divide between people who have access to and can use digital technologies and those who do not and ensuring people who lack access to digital technologies or the ability to use them do not fall through the cracks. Within the WHO European Region, 52% of Member States reported having digital health education policies or strategies.¹⁸ Additionally, 56% of the Member States' inclusion plans or strategies, like a digital literacy program, are aimed at marginalised communities.¹⁹ However, the adoption of these plans varies widely by subregion. As presented in **Figure 3**, all five central Asian Member States and eight out of the ten Member States in northern Europe reported digital inclusion plans, this was not the case in the remaining subregions where less than 50% of the responding Member States reported such plans. A similar uneven distribution

Figure 3: Digital health education action and digital inclusion plans

Note: For countries of the WHO European Region the geographic subregions are as defined by the United Nations Statistics Division and used in all United Nations publications and databases.

Source: ^{1,2}

of digital health education plans, policies, and strategies also differs notably across subregion.¹

The Digital Decade,³ which outlines the EU's vision for digital transformation, includes indicators of digital skills as some of its key performance metrics. According to the Digital Compass, by 2030, 80% of EU citizens between the ages of 16 and 74 should possess at least fundamental digital skills.⁴ However, the findings of WHO on digital health in the European Region show that less than 50 % of EU Member States have digital health education and inclusion plans.¹ To achieve this goal by 2030, these plans should be in place to ensure that we are bridging the digital divide and leaving no one behind.

Conclusion

The advancement and adoption of DHT across the WHO European Region offers promising avenues for more effective and accessible healthcare, particularly in the face of challenges like the COVID-19 pandemic. The growing adoption and utilisation of DHT such as patient portals, telehealth, and mHealth services reflects the region's commitment to leveraging technology to improve patient care.

However, alongside these advancements, persistent disparities in access and usage remain a concern. Factors such as geographical location, socio-economic status, age, and digital literacy contribute to the digital divide, potentially hindering the promise of UHC. Effective digital health governance and integrating digital and health literacy education into national health objectives is paramount. To bridge the digital divide, Member States should consider developing capacity-building and digital inclusion plans, policies, or strategies to ensure that everyone, regardless of background can access and effectively use DHT. These measures will accelerate action towards achieving UHC while leaving no one behind.

References

- 1 WHO Regional Office for Europe. *The ongoing journey to commitment and transformation: digital health in the WHO European Region*. Copenhagen: WHO Regional Office for Europe, 2023.
- 2 WHO Regional Office for Europe. *Equity within digital health technology within the WHO European Region: a scoping review*. Copenhagen: WHO Regional Office for Europe, 2022.
- 3 Eurostat. *Urban and rural living in the EU* [online database]. Brussels: European Commission, 2020. <https://ec.europa.eu/eurostat/web/products-eurostat-news/-/EDN-20200207-1>, accessed 20 September 2022.

⁴ van Kessel R, Wong BLH, Rubinić I, et al. Is Europe prepared to go digital? making the case for developing digital capacity: an exploratory analysis of Eurostat survey data. *PLoS Digital Health* 2022;1(2):e0000013.

⁵ WHO Regional Office for Europe. Regional Committee for Europe. Sixty-ninth Regional Committee for Europe: Copenhagen, 16–19 September 2019. *Resolution: towards the implementation of health literacy initiatives through the life course*. WHO, 2019. <https://iris.who.int/handle/10665/339662>

⁶ European Commission. *Europe's Digital Decade: digital targets for 2030*. Brussels: European Commission, 2023. https://commission.europa.eu/strategyand-policy/priorities-2019-2024/europe-fit-digital-age/europes-digital-decade-digital-targets-2030_en

⁷ United Nations Department of Economic and Social Affairs. Methodology. In: Statistics division [website]. New York: United Nations, 2023. <https://unstats.un.org/unsd/methodology/m49/>

Acknowledgement:

We would like to extend our sincere gratitude to the co-authors of “the ongoing journey to commitment and transformation: digital health in the WHO European Region, 2023” report: Elettra Ronchi, Andrea de Panizza, David Glance, Abhinav Devaria, and Stefan Buttigieg. We also thank our colleagues from the WHO Regional Office for Europe – Helen Caton-Peters, Ryan Dos Santos, Govin Permanand, and Clayton Hamilton – for their invaluable technical review and insights during the report's development. Our deep appreciation also goes to the expert contributors and survey respondents from all the Member States of the WHO European region for their invaluable contributions.